

SINGAPORE-MIT ALLIANCE

COMPUTATIONAL ENGINEERING

CME5232: Cluster and Grid Computing Technologies for Scientific Computing

COMPUTATIONAL LAB No. 1

5th July, 2006

INTRODUCTION TO HIGH PERFORMANCE COMPUTING RESOURCES

Objectives

To be familiar with the SMA Hydra Linux Cluster as the main High Performance Computing (HPC) platform for parallel computing tasks.

Background Knowledge

C/C++ programming languages, elementary knowledge of Unix/Linux commands.

CME5232 course information webpage

Some webpage containing useful course information (e.g. SMA Hydra Linux Cluster, etc) has been set up and its URL is as follows: **<http://web.sma.nus.edu.sg/sma5232/>**

Exercises (Due date: 12th July 2006) - You are required to submit only the homework problem. *Please submit a short formal report recording your observations based on the assigned exercises and programming assignment in a week's time. The report should be complete and clearly written to show what you have done and what you have observed and interpreted from these exercises. All equations, symbols etc. if any must be explained clearly.*

Sample programs can be found in the following Hydra directory **/home/cme5232/clab/clab1/**.

1. Getting to know the SMA Hydra Linux Cluster

You have been briefed on the HPC resources on the SMA Hydra Linux Cluster. By browsing through the hydra cluster website <http://hydra.sma.nus.edu.sg> (maintained by NovaGlobal Pte Ltd, <http://www.noveglobal.com.sg>), explore the cluster tools and language options to get a handle on the capabilities of the cluster.

2. The integration of a continuous line function can be approximated by using the trapezoidal rule. In this method, the area under the function is divided into a number of trapezoids of equal width. By adding up the areas of each trapezoid, the integration of the function can be estimated.

The function to be integrated is $f(x) = \frac{e^x}{x^3 + x + 1} + x^2 \sin x$ over the range $x \in [0, 10]$. Compile the sample serial program **ori_integral.c** and run the program with the following inputs.

$$a = 0, b = 10, \text{ vary } n \text{ from } 10^2 \rightarrow 10^7$$

Experiment with various optimization levels (O0, O1, O2 and O3) by invoking the compiler optimization and time the execution.

Homework Problem to Prepare for Computational Lab 2

In this problem, we are going to compute the value of **pi**. There are numerous ways of computing the value of **pi**. In this homework problem, students are required to choose either Monte Carlo method or Continuous method to evaluate the value of **pi**.

Monte Carlo method

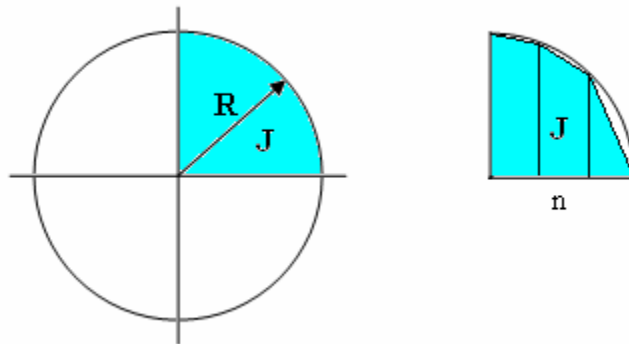
This method consists of inscribing a circle inside a square and then throwing random “darts” at the square. The ratio of “darts” that land inside the circle (**n**) to the total number of the darts thrown (**N**) will be the ratio of the areas of the circle to the square (**pi/4**). From this, the value of **pi** can be obtained as **pi = 4*n/N**.

In this lab, you are expected to write out the sequential version of the code.

Run your sequential code by varying the total number of the “darts” thrown from $N = 10^4$ to 10^8 and record the corresponding computation time. Record the value of **pi** and check the accuracy of your **pi** against PI25DT (PI25DT = 3.141592653589793238462643 is the true π value up to 25 digits of accuracy).

Continuous method

Consider a circle with radius **R**. Use the trapezoidal rule, as shown in question 2, to compute a quarter area of the circle, called **J**. Then the value of **pi** can be obtained as **pi = 4*J/R²**.



In this lab, you are expected to write out the sequential version of the code.

Run your sequential code by varying the number of trapezoids of equal width **n** from 10^3 to 10^8 and record the computational time. Record the value of **pi** and check the accuracy of your **pi** against PI25DT (PI25DT = 3.141592653589793238462643 is the true π value up to 25 digits of accuracy).